Fresher, Not Safer: Cross-Product SST Substitution Breaks a Percentile Anomaly Test in a Live Corroboration Pipeline

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Abstract. Scoring a fresh observation against a climatology fitted on a different data product is conventionally treated as a substitution that only affects timeliness. We report a case, measured directly inside a live marine anomaly-detection pipeline, where that convention is wrong. The system corroborates a wind/currents-derived upwelling index against an independent cold sea-surface-temperature (SST) signal, scored two ways against the same climatology: a weak, mean-based test (a fixed 0.5°C anomaly threshold) and a strong, percentile-based test (falling below the seasonal 10th percentile), each reported as a contrast between upwelling-favourable and control coastal cells; 2 independently attempted fixes for the same latency-driven product mismatch — swapping the scored SST source to a fresher, higher-resolution live product (Fix A, measured 2026-08-17), and fitting a matched climatology on that same product (Fix B, measured 2026-08-25) — both leave the weak tier's contrast nearly unchanged while inverting the strong tier's sign, from a near-zero matched-arm baseline to -0.149 and -0.051 respectively. A direct one-day per-cell comparison (2026-08-01) locates the cause: no systematic offset between the two products, but per-cell dispersion of 0.467°C standard deviation in open water, rising to 0.758°C across the coastal band — wider than the weak tier's own threshold and comparable in scale to the climatological gap the percentile boundary sits close to, which is why the same noise degrades one test gracefully and destroys the other. A third, controlled measurement (Control C, 2026-08-28) rules out an alternative explanation by holding the SST product and its climatology fixed and varying only the wind/currents input's temporal aggregation instead: both tiers move only slightly, in the same direction, and by less than the matched arm's own run-to-run variation — evidence against instantaneous sampling being the cause. Every number reported here is a single, dated, live measurement of a moving field, not a standing artifact, and the general rule this evidence points toward — safe substitution up to a threshold set by the inter-product spread — remains to be established beyond the two source pairs measured here.

Keywords: sea surface temperature, percentile anomaly detection, marine heatwave, climatology, cross-product bias, corroboration test, upwelling, reproducibility

1 Introduction

Anomaly detection over a geophysical field almost always needs two things that rarely come from the same place: a long historical record to fit a climatological baseline on, and a fresh observation to score against it. A satellite composite with a decade or more of consistent processing is the natural choice for the first; whatever product a pipeline already has running live — a different satellite, a reanalysis, a numerical-model nowcast — is the natural choice for the second. Swapping the fresh observation's source for something newer, higher-resolution, or lower-latency than the product the baseline was fitted on is, by the convention this practice runs on, a strict improvement: it reads as closing a timeliness gap and nothing else. This paper reports a case, measured directly inside a running platform, where that convention is wrong, and wrong in a way that a mean-based diagnostic does not reveal.

The system is MarisAI's upwelling detector, which corroborates a wind/currents-derived upwelling-favourability index against an independent cold sea-surface-temperature (SST) signal: water that a favourable wind stress should be pulling up from depth ought to also read anomalously cold. That corroboration is scored two ways against the same climatological baseline — a weak, mean-based test (the observed anomaly from the climatological mean exceeds a fixed 0.5°C) and a strong, percentile-based test (the observation falls below the seasonal 10th percentile, a tail test) — and both are reported as a *contrast*: the fraction of favourable coastal cells passing the test minus the same fraction on downwelling-favourable cells, where no cooling signal is physically expected. Two independent attempts to close OISST's inherent latency disadvantage against a fresher, live product both leave the weak tier of that contrast nearly unchanged while inverting the strong tier — not by a small amount, and not the same amount each time, but always in the failing direction. A third, controlled measurement, run to rule out an alternative explanation for the first two rather than as a third attempted fix, confirms that the cause is the product substitution itself and not an artifact of scoring an instantaneous snapshot.

1.1 Contributions

1. A **live, measured demonstration** that scoring an observation against a climatology fitted on a different product is not a latency-only substitution: 2 independently attempted fixes for the same underlying mismatch — swapping to a fresher, higher-resolution live product (§4, Fix A), and fitting a matched climatology on the product actually being scored (Fix B) — both invert the strong, percentile-based tier of a real operational corroboration test while leaving its weak, mean-based tier close to unchanged.
2. A **direct measurement of the mechanism**: one day of paired observations from both products, coarsened onto the same grid, shows no systematic offset (both medians near zero) but per-cell dispersion of 0.467°C standard deviation in open water, rising to 0.758°C across the coastal band — wider

than the 0.5°C threshold the weak tier is built around, and wide enough to randomise membership in a percentile tail.

3. A **control measurement that rules out the obvious alternative explanation**: holding the scored product and its climatology fixed and varying only the wind/currents input's temporal aggregation (an instantaneous snapshot against a trailing mean) leaves both tiers of the contrast close to zero, which is evidence *against* instantaneous sampling being the cause of the two inversions above, not a third instance of the same failure.
4. An explicit **evidence grading** of every number reported: every measurement here is a single, dated, live run against a moving field, with no stored output file committed to the repository, and this paper states that plainly rather than presenting any of it as a standing fact.

2 Related Work

Percentile-based anomaly detection as domain convention. The mean-versus-percentile distinction this paper's failure mode turns on is not an implementation detail specific to this platform; it is how the field already defines a marine heatwave. Hobday et al. [1] establish the now-standard hierarchical definition: an SST anomaly is categorised by how far it sits above the seasonally varying 90th (warm) or below the 10th (cold) percentile of a fitted climatological baseline, precisely the tail-test construction §3 describes for the cold-anomaly corroboration case. That definition is widely adopted exactly because it is more diagnostic of an unusual event than a fixed absolute threshold on the mean anomaly — the same property that, this paper shows, makes it more fragile to a source of noise a mean-based threshold tolerates.

The scored and fitted products themselves. The baseline this platform's production corroboration uses is NOAA's Optimum Interpolation SST, whose current version is documented by Huang et al. [2]; the live product substituted for it in Fix A, and the reanalysis substituted in Fix B, are both from the Copernicus Marine Service's global physics system. Reynolds and Chelton [4] compare multiple independently produced daily SST analyses directly and find measurable disagreement between them at exactly the spatial scale — coastal, sub-degree structure — this paper's own per-cell comparison locates the largest disagreement in. That result establishes that SST products disagree by a measurable amount; it does not, on its own, connect that disagreement to what happens to a downstream percentile test that consumes one product's climatology and another's fresh observation. That connection, quantified against a real operational corroboration test rather than as a general property of the products, is this paper's contribution.

Companion catalogue. This finding was first catalogued as one of eleven silent, plausible, signed-direction failure instances found across this same platform [3], where it is instance I.12, summarised in two sentences with the same headline

numbers as §4's Fix A and Fix B. This paper is the full methodological treatment of that single instance: the corroboration-test design, the per-cell disagreement measurement, both fix attempts, and the control measurement that catalogue does not include.

3 Methods

The climatology. `services/climatology/build.py` fits a per-cell, per-day-of-year climatology: a mean and the 10th and 90th percentiles, pooled over a ±5-day window around each calendar day across every year in a baseline period, following the field's own day-of-year percentile convention [1]. Two such climatologies exist in this codebase. The production one is fitted on the OISST daily record over 1991–2020. A second was built specifically to test Fix B, fitted on the Copernicus GLORYS reanalysis (`cmems_mod_glo_phy_my_0.083deg_P1D-m`) over 1993–2022 — not the same 30-year window as OISST's, because the reanalysis's own coverage does not reach back past 1993; `scripts/build_climatology_copernicus.py` states this explicitly and would refuse a 1991 start outright rather than silently truncate it.

The scoring shape. `services/sst_anomaly.py` defines the object both consumers score against: for a given observation and climatology, an `anomaly` field (observed minus the day's climatological mean) and a `cold_exceedance` field (observed minus the day's 10th percentile; negative means below it). The weak tier tests `anomaly` ≤ -0.5°C, a fixed absolute threshold (`services/upwelling.py`'s `COOL_ANOMALY_C`); the strong tier tests `cold_exceedance` < 0, the percentile tail.

The corroboration test and its control arm. `services/upwelling.py` computes a wind/currents-derived upwelling-favourability index per coastal cell and corroborates it against the SST signal above. Following this same platform's own established discipline that a corroboration rate is not evidence of causation without a control arm [3], every result below is reported as a *contrast*: the fraction of upwelling-favourable cells passing a given tier's test, minus the same fraction on downwelling-favourable cells, where a cold anomaly is not physically expected. A positive contrast is the corroboration signal; a contrast near zero on the matched, production configuration is the expected baseline this paper's fix attempts are measured against.

Fix A — swap the scored product. Measured 2026-08-17. `services/copernicus_sst.py`'s `anomaly_field()` was, at this date, implemented to score the live, hourly Copernicus physics field — the same field cached for the platform's map layer, hours old against OISST's own ∼14.5-day publication lag — against the OISST-fitted climatology: a fresher, higher-resolution observation, scored against a baseline fitted on a different

product. Both this configuration and the production OISST-on-OISST configuration were scored against the identical live wind/currents snapshot, so the only variable between the two arms of Table 1's first row is the SST source.

The per-cell disagreement measurement. Measured 2026-08-01, independent of the corroboration test above: one day of hourly Copernicus physics, daily-averaged and coarsened onto the OISST grid, compared cell-by-cell against OISST's own value for that day, separately for open-ocean and coastal cells.

Fix B — fit a matched climatology on the product being scored. Measured 2026-08-25, after Fix A's failure diagnosed the cause as product mismatch rather than latency. `scripts/build_climatology_copernicus.py` built the GLORYS-fitted climatology described above, and `services/copernicus_sst.py`'s `anomaly_field()` was repointed to it (`REANALYSIS_CLIMATOLOGY_VARIABLE`) — the version shipped in the code today, though the production corroboration in `services/upwelling.py` still consumes `heatwaves.sst_anomaly_field()` (OISST-on-OISST), not this path. Both arms of Table 1's second row are reproduced by `scripts/measure_sst_corroboration.py --source {oisst, copernicus_reanalysis}`, again over the identical live wind/currents snapshot on the measurement date.

Control C — vary temporal aggregation, hold the product matched. Measured 2026-08-28. This is not a third attempted fix for the product mismatch; both arms score OISST against its own OISST-fitted climatology throughout. What varies is the wind/currents side of the corroboration: `services/wind_history.py` keeps a trailing mean of Ekman transport, populated for this measurement by backfilling 30 real historical Copernicus wind timesteps (`scripts/backfill_wind_history.py`, spanning 1.2 days) rather than the 3-day window originally targeted, because that many real timesteps were what the backfill could obtain. An instantaneous snapshot (`upwelling.detect`) is compared against the trailing mean (`upwelling.detect_from_history`) via `scripts/measure_wind_history_corroboration.py`. The purpose is to rule out an alternative explanation for Fix A and Fix B's inversions — that scoring an instantaneous wind/currents snapshot against an instantaneous SST reading, rather than the product mismatch itself, is what destabilises the tail test.

Evidence grade. Every number in §4 is a single, live measurement of a moving field — a day's wind, currents, or SST state — named with its date and the script that reproduces it. The four named scripts (`measure_sst_corroboration.py`, `build_climatology_copernicus.py`, `measure_wind_history_corroboration.py`, `backfill_wind_history.py`) and the module they exercise (`services/wind_history.py`) are committed to the repository and were confirmed present; no stored output file, report, or

regression test backs any of the headline numbers. Following this project's own reproducibility ledger convention [3], this is graded *Dated* throughout — reproducible on demand from a committed script, but not a standing fact about the current state of any field, and not backed by a committed artifact.

4 Results

Table 1. Two independently attempted fixes for the same product mismatch, each inverting the strong, percentile-based tier while leaving the weak, mean-based tier close to unchanged. Control C holds the product and climatology matched throughout and varies only the wind/currents input's temporal aggregation; it is not a third fix attempt (see §5). All contrasts are favourable minus control (downwelling-favourable) coastal cells, over the identical live wind/currents snapshot within each row.

Attempt (date)	Arm	Weak contrast	Strong contrast
Fix A: swap product (2026-08-17)	OISST (matched)	0.026	0.002
	Live physics (mismatched)	0.022	-0.149
Fix B: matched climatology (2026-08-25)	OISST (matched)	0.027	-0.001
	GLORYS-fitted (matched)	0.021	-0.051
Control C: wind aggregation (2026-08-28)	Instantaneous	0.0234	-0.0061
	1.2-day trailing mean	0.0182	-0.0049

Fix A inverts the strong tier. Swapping the scored observation from OISST to the live physics field, holding the climatology fixed, leaves the weak tier essentially unchanged (0.026 against 0.022) but moves the strong tier from a near-zero 0.002 to -0.149 — downwelling-favourable (control) coastal cells came out below their seasonal 10th percentile roughly three times as often as favourable ones, the opposite of what the test is supposed to show.

The cause is dispersion, not offset, and it concentrates on the coast. The paired one-day comparison shows no systematic bias to correct: median disagreement is 0.011°C in open water and -0.002°C across the coastal band — both close to zero, so a constant bias term would be a fudge with nothing to fix. What differs sharply between the two is dispersion and mean: open-ocean cells show a mean disagreement of 0.033°C and standard deviation 0.467°C, with 14.3% of cells differing by more than 0.5°C and 3.9% by more than a full degree; the coastal band is worse on every one of those figures — mean 0.131°C, standard deviation 0.758°C, 24.9% and 10.0% respectively. The coastal standard deviation alone is wider than the 0.5°C threshold the weak tier is built around: a percentile tail

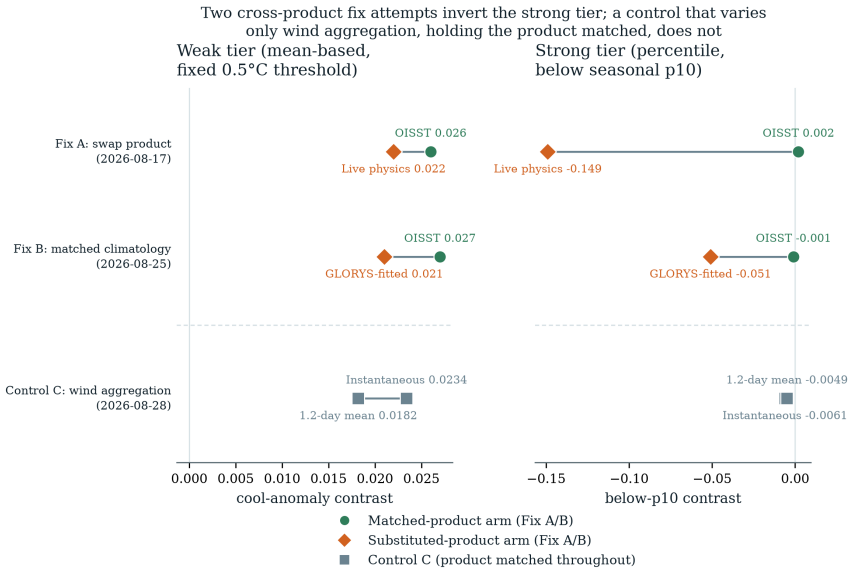


Fig. 1. The same six rows as Table 1, plotted as the arm each attempt compares against its matched baseline. Left: the weak, mean-based tier barely moves in any of the three attempts. Right: the strong, percentile-based tier inverts sign in Fix A and Fix B (the two cross-product substitutions) but stays near zero, same sign, in Control C (product held matched throughout) – visually separated from the first two rather than grouped with them, because it answers a different question.

test, whose membership condition already sits close to a base rate of $100 - 10\%$ by construction, cannot survive noise the width of its own signal, while a fixed-threshold test tolerates the same noise because the threshold is large relative to the noise's mean effect on either side of the comparison.

Fix B does not rescue the tail test. Fitting the climatology on the same product family as the live observation removes the cross-product disagreement term measured above by construction — the GLORYS-fitted climatology and the live Copernicus physics field share a processing chain. It does not rescue the strong tier: scored the same way on the same day, the matched OISST arm reads 0.027 weak / -0.001 strong, and the matched GLORYS arm reads 0.021 weak / -0.051 strong — the weak tier again barely moves, and the strong tier again inverts, to a different but still substantially negative value. Removing the specific 0.758°C-scale disagreement Fix A's diagnosis pointed at was not sufficient; whatever destabilises the tail test survives fitting a matched baseline.

Control C rules out instantaneous sampling as the explanation. Holding both the product and the climatology fixed at OISST-on-OISST throughout and varying only the wind/currents field's temporal aggregation moves both tiers, but only slightly, and in the same direction on both sides: weak 0.0234 to 0.0182, strong -0.0061 to -0.0049, a movement of 0.0012 on the strong tier. That is smaller than the 0.003 the matched arm's own strong-tier value moved between Fix A's measurement date and Fix B's — run-to-run variation the matched configuration shows on its own, with nothing about the wind input changed at all. Control C's contrasts stay the same sign and close to zero in both arms; nothing in this measurement resembles Fix A's or Fix B's inversion. This is evidence against instantaneous sampling being what destabilises the tail test, which leaves the cross-product substitution itself, demonstrated twice under two different attempted repairs, as the explanation the data actually supports.

5 Discussion

Three measurements, two failures and one control, not three failures. It would overstate this result to describe Control C as a third instance of the same inversion: its own numbers do not invert, and its own reproduction script states directly that it holds the SST source fixed at the one production actually uses, varying only the wind. Its role is to close off the most obvious objection to reading Fix A and Fix B as evidence of a genuine product-mismatch effect rather than an artifact of scoring one instantaneous field against another — and by moving the contrast only slightly, in the same direction on both arms, and by less than the matched arm's own run-to-run wander, it does close that off. The honest structure of this paper's result is two independently attempted, different-mechanism fixes for the same underlying mismatch failing in the same qualitative way, plus a control that rules out the leading alternative explanation for why.

Why a tail test is more fragile than a mean-offset test. A fixed absolute threshold on a mean-referenced anomaly and a seasonal percentile threshold are both, in principle, ways of asking "is this cell unusually cold." They differ in how close their decision boundary sits to the distribution of plausible values. The weak tier's 0.5°C threshold is large relative to the corroboration signal being measured (a contrast on the order of a few hundredths); injected, near-zero-median noise of the measured dispersion perturbs cells near that boundary on both the favourable and control sides by a similar amount, so the *contrast* — already a difference of two small fractions — moves little. The strong tier's boundary is the seasonal 10th percentile, a threshold defined to sit close to the edge of the baseline distribution by construction; the coastal-band dispersion measured here (0.758°C) is comparable in scale to the climatological gap between the mean and that percentile in the first place, so the same injected noise can move a much larger share of cells across the boundary, in either direction, than a mean-based threshold ever notices. This is a property of the boundary's own tightness relative to the noise floor, not of either product being wrong: OISST and the Copernicus physics products are each defensible, quality-controlled estimates of the same physical quantity, and the disagreement between them is ordinary product-to-product variation of the kind the wider SST literature already documents [4]. What is not ordinary, and not previously connected to that literature in this form, is that the same variation degrades one class of downstream test gracefully and destroys the other.

A practical recipe. The two options this paper's own evidence supports are not equally well demonstrated. Quoting the inter-product spread beside the threshold — reporting, alongside any percentile-based anomaly claim, the dispersion between the product the climatology was fitted on and the product the fresh observation actually came from — costs one day of overlapping data and is directly what §4's per-cell comparison already does. Fitting the baseline on the product actually being scored is the more complete fix in principle, but this paper's own Fix B measurement is a demonstration that it is *not sufficient on its own*: the matched-product climatology removed the specific disagreement term diagnosed after Fix A and the tail test still inverted. The recipe this result supports is therefore the first option as a minimum bar for any percentile-based claim scored across a product boundary, with matching the baseline product as a partial mitigation whose own residual risk should be re-measured, not assumed away, exactly as this paper had to do a second time after Fix B did not close the gap Fix A opened.

6 Limitations and Future Work

Every number in this paper is a single, dated, live measurement of a moving field, and none is backed by a committed report or a passing regression test; §3's evidence grade says so explicitly and this section repeats it because a result built entirely on dated numbers should not be quietly read as more solid than that.

Each of the three measurements is one snapshot, not a repeated-sampling result: Control C's own docstring is explicit that "this one run is a single snapshot at a 1.2-day window, not a repeated-sampling result," and does not rule out the originally targeted 3-day window or a different sampling scheme, only the specific window the real backfill happened to obtain. The infrastructure to re-ask that question — `wind_history.py`, `detect_from_history`, and the backfill script — is already built, which makes a longer window a re-run rather than a re-build, but that re-run has not been done as of 2026-09-09.

This paper's scope is one region (the platform's coastal domain), one variable (SST), and two source pairs (OISST/live-Copernicus-physics, OISST/GLORYS-reanalysis). The general form this evidence points toward — that the substitution is safe up to a threshold approximately proportional to the inter-product standard deviation, so a rule of the shape "safe below threshold $\approx k\times$ inter-product sd" could be stated and tested — has not been established here and is not established anywhere in this platform's own research notes as of 2026-09-09; it remains exactly the open item those notes already name, repeating this design across two or three further product pairs and both tiers rather than the two pairs and the incidental control measured here. Nothing in either attempted fix, or in the control measurement, supersedes that open item; both fix attempts add a second and third dated data point to the same single-region, single-variable case, not a generalisation across it.

7 Conclusion

Scoring a fresh observation against a climatology fitted on a different data product is treated, by the convention this practice runs on, as a substitution that only affects timeliness. Measured directly inside a live operational corroboration test, it is not: two independently attempted fixes for the same underlying product mismatch — swapping to a fresher, higher-resolution live product, and fitting a matched climatology on the product being scored — both leave a weak, mean-based test of the same signal nearly unchanged while inverting its strong, percentile-based tail test, and a control measurement that varies only the wind input's temporal aggregation rules out instantaneous sampling as an alternative explanation for either. The cause is per-cell dispersion between the two products wider than the decision threshold the weak tier tolerates and comparable in scale to the climatological gap a percentile boundary is built close to. Every number here is a dated, single measurement of a moving field, not a standing fact, and the general rule this evidence points toward — safe substitution up to a threshold set by the inter-product spread — remains to be established across more than the two source pairs measured here.

Reproducibility

Every number in this paper is graded *Dated* in §3: a single, live measurement of a moving field, reproducible on demand from a committed script, with no

stored output file backing any headline number. Fix A and Fix B are reproduced by `scripts/measure_sst_corroboration.py --source {oisst, copernicus_reanalysis}`; the GLORYS-fitted climatology Fix B scores against is built by `scripts/build_climatology_copernicus.py`; the per-cell disagreement measurement is read directly from `services/sst_anomaly.py`'s own module docstring. Control C is reproduced by `scripts/measure_wind_history_corroboration.py --backfill-hours 72`, backfilling real wind history via `scripts/backfill_wind_history.py` into `services/wind_history.py`.

All five files are committed to the repository and were confirmed present as of 2026-09-09; none of their outputs is.

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